

Training-Path Dependence and Conservation Reliability in Heterogeneous Groundwater PINNs

An exploratory numerical research guide

Abstract

We investigate Physics-Informed Neural Networks from a manufactured, one-dimensional Poisson benchmark towards a synthetic, smooth heterogeneous groundwater-flow PDE. Six phases evaluate conductivity contrasts, training durations, fixed and adaptive boundary loss weights, mechanism ablations, gradient alignment and checkpoint-intervention timing. The observed rankings depend on optimization history and the evaluation metric; boundary fit, head error, and flux conservation should be assessed separately.

1. Motivation and prior work

The main starting points were Cuomo et al. (2022), Scientific Machine Learning through Physics-Informed Neural Networks: Where We Are and What’s Next (DOI: 10.1007/s10915-022-01939-z). Section 5 emphasizes loss design, convergence, sampling, numerical stability and challenging physical problems, while Wang, Teng and Perdikaris (2021; DOI: 10.1137/20M1318043) provide context for gradient-flow pathologies. Esmailbeigi et al. (2026; DOI: 10.1007/s10915-026-03447-w) contextualize established loss-weighting methods. We did not re-implement the algorithms published by these authors.

2. Mathematical models and measurements

Poisson:

$$-u''(x) = \pi^2 \sin(\pi x) + 0.2(3\pi)^2 \sin(3\pi x), \quad (1)$$

with zero Dirichlet conditions and exact solution

$$u(x) = \sin(\pi x) + 0.2 \sin(3\pi x). \quad (2)$$

Groundwater:

$$-(k(x)h'(x))' = 1, \quad h(0) = 1, \quad h(1) = 0, \quad (3)$$

and

$$k(x) = \exp\left[-\log(C) \operatorname{sigmoid}\left(\frac{x - 0.5}{0.04}\right)\right], \quad C \in \{1, 10, 100\}. \quad (4)$$

This is a smooth synthetic representation of conductivity, not a real-world geological interface. Darcy flux $q = -kh'$, so that $q' = 1$. The reference solution is obtained by numerical integration of this identity.

Loss:

$$L = L_{\text{PDE}} + \lambda L_{\text{BC}}. \quad (5)$$

The adaptive heuristic periodically balances gradient norms by varying λ toward clipped

$$\frac{\|\nabla L_{\text{PDE}}\|}{\|\nabla L_{\text{BC}}\| + \varepsilon}. \quad (6)$$

Evaluation includes relative L^2 error in head, boundary mismatch, error in flux profile, conservation RMSE, and cosine similarity of gradient vectors in PDE and BC parameters.

3. Phase-by-phase design, recorded results and interpretation

Phase 1: Poisson baseline

12 paired seeds, 800 Adam epochs, fixed weight of 1 versus gradient-ratio adaptive method. Median relative L2: 0.001655 vs 0.000142. The adaptive method looked better than one unfitted baseline; this is not evidence of optimality.

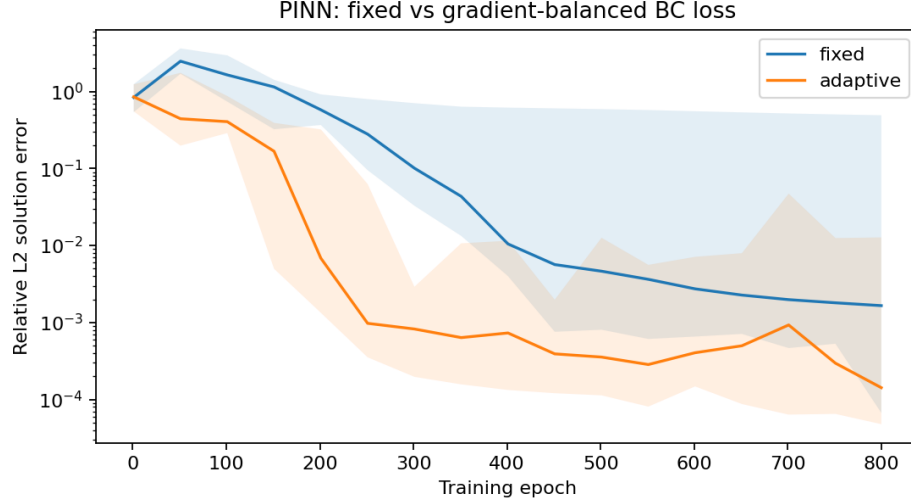


Figure 1: Recorded figure from archived experimental outputs; not a newly rerun experiment.

Phase 2: Conductivity contrast

$C=1,10,100$; 18 total runs, 1200 epochs. $C=100$ median head errors: 0.142405 (fixed 1) and 0.140363 (adaptive). Increasing heterogeneity made fitting harder under this training budget.

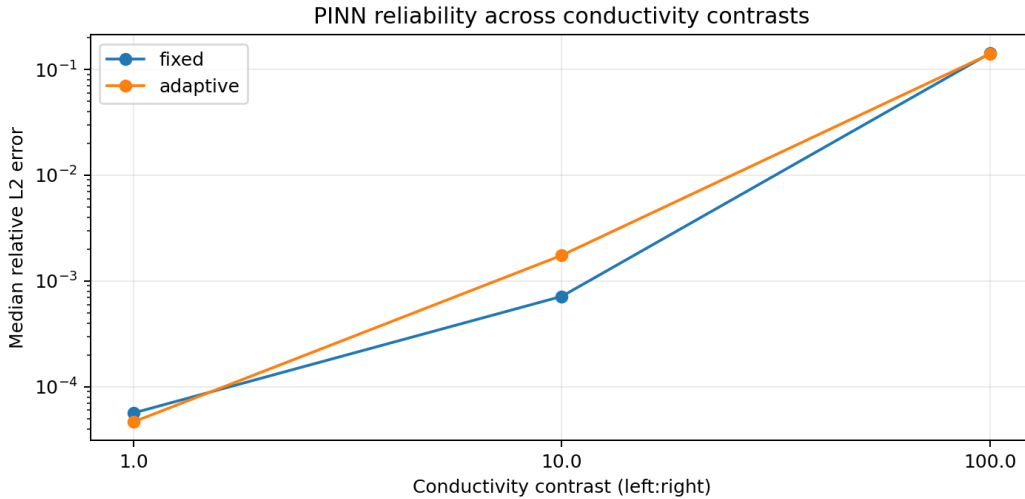


Figure 2: Recorded figure from archived experimental outputs; not a newly rerun experiment.

Phase 2b: Training-budget diagnostic

$C=100$, 5000 epochs, two seeds. Fixed errors 0.005747 and 0.001248, adaptive 0.020584 and 0.001828. The earlier error did not establish an intrinsic inability to represent the solution.

Phase 3: Fixed-weight sweep

C=100, 2500 epochs, three seeds. Median head L2 and conservation RMSE: fixed 0.1 = 0.002759 / 0.034207; fixed 1 = 0.012710 / 0.069096; adaptive = 0.06581 / 0.15053. A tuned fixed baseline exceeded our heuristic.

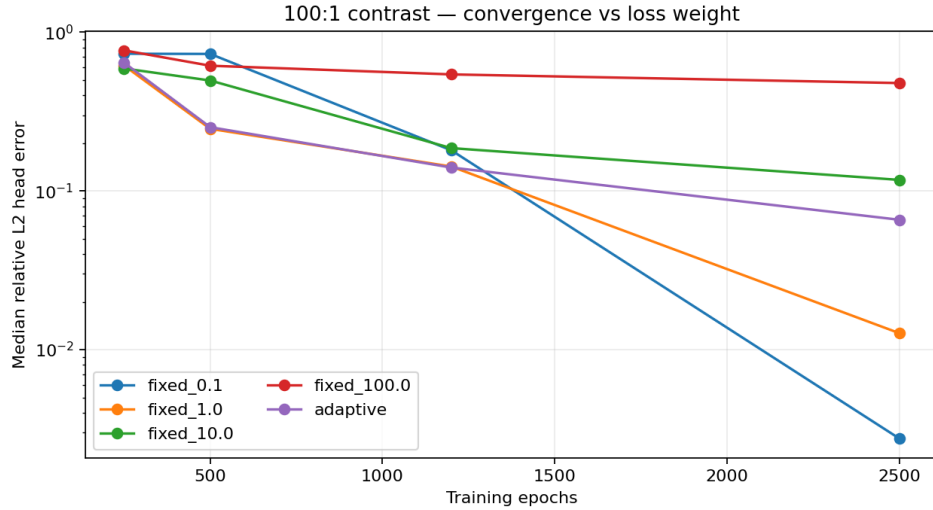


Figure 3: Recorded figure from archived experimental outputs; not a newly rerun experiment.

Phase 4: Adaptive-weight cap ablation

At C=100, capping lambda at 1 improved median head L2 to 0.016673 relative to original adaptive 0.06581. Suggests overweighting may contribute; cap at 0.1 is equivalent to fixed 0.1 by construction.

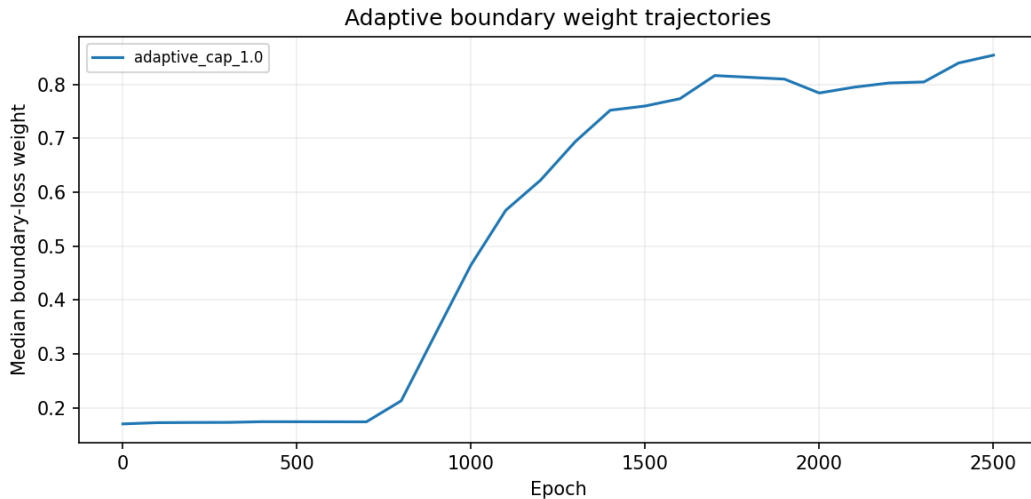


Figure 4: Recorded figure from archived experimental outputs; not a newly rerun experiment.

Phase 5: Identical-state intervention

Pretrain 500 epochs at lambda=1, copy model and Adam state, branch another 1200. Medians: head L2 0.10057 (fixed 0.1), 0.07654 (fixed 1), 0.09624 (adaptive). Conservation RMSE 0.16262, 0.17453, 0.19254 respectively. Starting state matters.

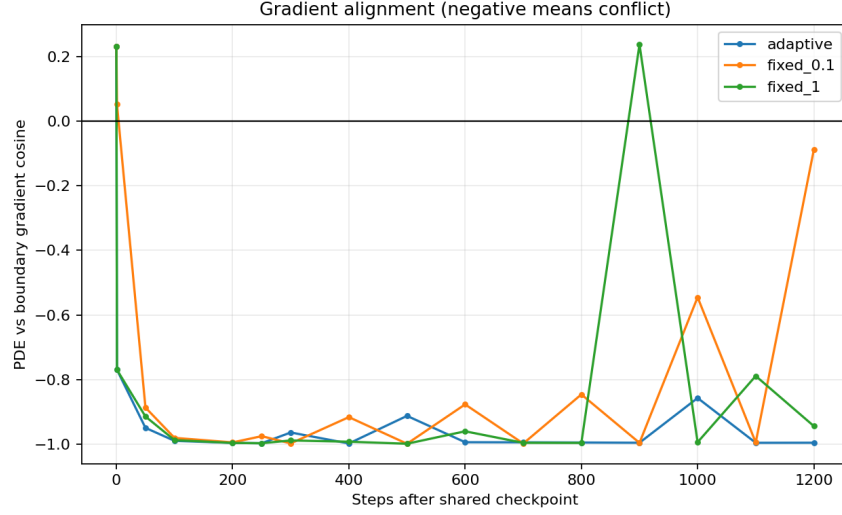


Figure 5: Recorded figure from archived experimental outputs; not a newly rerun experiment.

Phase 6: Timing experiment

Ten paired seeds, three pretraining checkpoints 250/1000/2500, then 400-step branches. Fixed 1 gives lowest median head error at all checkpoints: 0.17745, 0.13743, 0.00961. Flux rankings differ. Inter-checkpoint comparisons have unequal total training budgets.

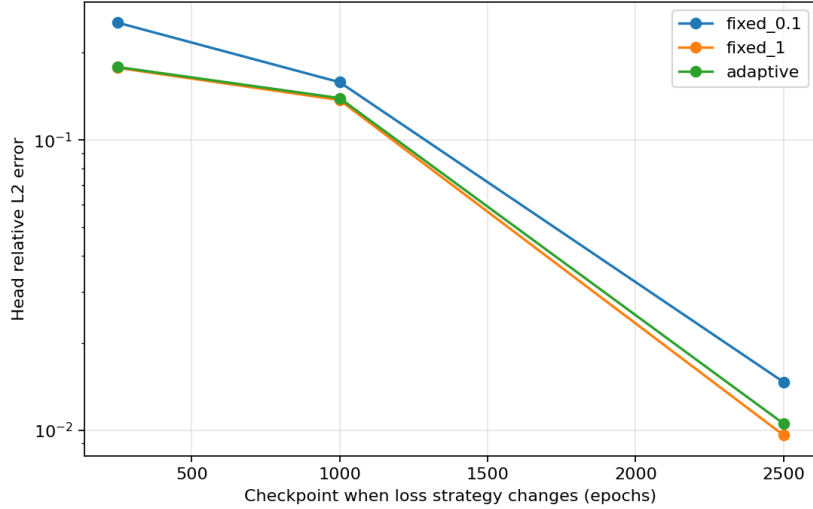


Figure 6: Recorded figure from archived experimental outputs; not a newly rerun experiment.

4. Ten-seed timing-study numeric evidence

Median hydraulic-head relative L^2

Checkpoint	Fixed 0.1	Fixed 1	Adaptive
250	0.25394	0.17745	0.17857
1000	0.15832	0.13743	0.13953
2500	0.01465	0.00961	0.01053

Median flux conservation RMSE

Checkpoint	Fixed 0.1	Fixed 1	Adaptive
250	0.30006	0.26173	0.26243
1000	0.21664	0.23720	0.23777
2500	0.06203	0.06595	0.06156

Median PDE/BC gradient cosine

Checkpoint	Fixed 0.1	Fixed 1	Adaptive
250	-0.93779	-0.80443	-0.96241
1000	-0.97517	-0.99169	-0.99527
2500	-0.412	-0.284	-0.808

5. Interpretation and competing explanations

The studies suggest sensitivity to loss scaling, training budget, and model/optimizer trajectory. Opposing gradients often happen, but cosine similarity does not predict head accuracy. Boundary-weight inflation can lead to unfavorable trade-offs, but the checkpoint studies do not uniquely identify it as the cause. Flux-conservation performance can favor a different strategy than head error.

6. Limitations and threats to validity

Synthetic smooth 1D conductivity only; no real aquifer data or truly discontinuous interfaces. Three seeds in most studies and 10 only in the final timing study; no broad statistical generalization. Changing epoch budgets and matching only gradient-step counts does not match wall-clock compute. The adaptive heuristic is not a benchmark of established published adaptive optimizers. Fixed penalties were only partially tuned. Numerical differentiation used in conservation evaluation adds discretization error. No theorem proves PINN convergence, and no new method is claimed.

7. Reproducibility: how the companion Colab is used

1. Open Guided_PINN_Groundwater_All_Phases_Colab.ipynb in Google Colab.
2. Run dependency/setup cell.
3. Upload research_evidence.zip when prompted.
4. Follow the phase-order table/figure inspection cells.
5. Use the original code display and optional execution cells to reproduce phases.
6. Run the original contrast notebook separately.
7. Keep new output separate from archived results; compare software versions, random seeds, physical parameters and training budgets.

8. Next research directions

A controlled follow-up should compare matched wall-clock budgets and matched total optimizer steps, including hard Dirichlet boundary enforcement $h_\theta = 1 - x + x(1 - x)N_\theta(x)$. Test

truly discontinuous conductivity using interface-aware residuals with flux continuity, additional conductivity contrasts and transition widths, and correctly implemented published adaptive schemes. Use paired confidence intervals, more seeds, regional residual diagnostics and robust mass-balance checks. Longer-term extend to two dimensions and sparse-data inverse hydraulic-conductivity estimation.

9. Literature and technical resources

Cuomo et al. (2022), DOI 10.1007/s10915-022-01939-z; Wang, Teng & Perdikaris (2021), DOI 10.1137/20M1318043; Esmailbeigi et al. (2026), DOI 10.1007/s10915-026-03447-w. Actual experiment tools: PyTorch autograd/Adam; Python; NumPy; pandas; Matplotlib; Colab-compatible notebooks. No downloaded geology dataset or measured groundwater records were used.

For the complete verbatim code, see the seven source entries and original notebook bundled in `research_evidence.zip`. All reported numbers are from archived experimental artifacts; the PDF itself does not claim new compute was performed.